INCH-POUND

A-A-59754
21 October 2004

COMMERCIAL ITEM DESCRIPTION

GENERATOR, SIGNAL
(10 MHz to 20 GHz)

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This Commercial Item Description, (CID), describes a 10 MHz to 20 GHz a synthesized microwave source with a Continuous Wave (CW), Analog, Alternate, Manual, and List Sweep Modes. The equipment shall be capable of providing Amplitude Modulation (AM), Frequency Modulation (FM), and Pulse Modulation (PM). Only manufacturers meeting or surpassing the following requirements are to be supplied per this CID.

2. CLASSIFICATION. The equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet this CID. This equipment shall be Class 3 in accordance with MIL-PRF-28800.

3. SALIENT CHARACTERISTICS. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

3.1 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

3.2 Electrical Power Sources. The equipment shall be capable of operating from commercial, military, and shipboard power sources of 110/220 volts AC (±10%) and at line frequencies of 50 Hz to 440 Hz. The power consumption of this equipment shall not exceed 500 VA while operation from any power source specified herein. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein.

3.3 Reliability. Unless otherwise specified, the design of the equipment shall be such that under normal use and operation the equipment does not fail within 8000 hours of operation with a statistical certainty of 95%.

3.4 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater, based on an operating time of 2000 hours per year.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611

FSC 6625

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3.5 Performance Characteristics.

3.5.1 Frequency Range. The instrument shall be capable of a frequency range of 10 MHz to 20 GHz.

3.5.1.1 Resolution. The instrument shall be capable of a frequency resolution of 1 kHz.

3.5.1.2 Aging. The instrument shall have an aging rate of ≤ 2 x 10^{-8}/day, within the operating temperature range of the instrument.

3.5.2 Modes of Operation.

3.5.2.1 Continuous Wave (CW) Mode. The synthesizer shall have the following CW mode characteristics.

3.5.2.1.1 External 10 MHz Reference Input. The synthesizer shall accept external 10 MHz ±100 Hz, -10 to +20 dBm time base signal.

3.5.2.1.2 10 MHz Reference Output. The synthesizer shall be capable of 0 dBm from the reference output into 50 Ω AC coupled.

3.5.2.1.3 Switching Time. The instrument shall be capable of a switching time of ≤ 50 ms to be within 1 kHz of the final frequency.

3.5.2.2 Analog Sweep Mode. The instrument shall be capable of the follow analog sweep mode characteristics.

3.5.2.2.1 Sweep Width. The sweep width shall be independently selectable from 1 MHz to full frequency range.

3.5.2.2.2. Accuracy. The Analog Sweep Mode shall be capable of the minimum accuracy of ± 30 MHz for sweep speeds of ≤ 50 MHz/ms.

3.5.2.2.3 Sweep Time Range. The Analog Sweep Mode shall be capable of a sweep time range of 30 ms to 99 seconds.

3.5.2.3 List Sweep Mode. The instrument shall be capable of operation from a stored list of sweep data.

3.5.2.4 Spectral Purity.

3.5.2.4.1 Harmonics

Frequency Range	Standard (dBc)
≥ 10 MHz to ≤ 50 MHz	≤-30
> 50 MHz to ≤ 2 GHz	≤-40
> 2 GHz to ≤ 20 GHz	≤-60

3.5.2.4.2 Non-harmonics

Frequency Range	Standard (dBc)
≥ 10 MHz to ≤ 2 GHz	≤-40
> 2 GHz to ≤ 20 GHz	≤-60

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3.5.2.4.3 Single Sideband (SSB) Phase Noise. The instrument shall be capable of the following maximum acceptable parameters.

	Offset from carrier(dBc/Hz)			
Frequency Range	100 Hz	1 kHz	10 kHz	100 kHz
≥ 10 MHz to ≤ 2 GHz	-77	-88	-86	-100
≥ 2 GHz to ≤ 6 GHz	-78	-88	-86	-102
> 6 GHz to ≤ 10 GHz	-73	-86	-83	-102
> 10 GHz to ≤ 20 GHz	-66	-78	-78	-100

3.5.2.4.4 AM Noise Floor. The instrument shall have an AM noise floor of –130 dBm/Hz or less, with 0 dBm output and offsets ≥5 MHz from carrier.

3.5.2.5 RF Output. The instrument shall be capable of the following characteristics.

3.5.2.5.1 Residual FM. The minimum acceptable parameter shall be ≤ 120 Hz, RMS for frequencies ≤ 10 MHz to < 8 GHz, and ≤ 220 Hz, RMS for frequencies ≥ 8 GHz to ≥ 20 GHz.

3.5.2.5.2 Maximum Leveled Output Power. The minimum acceptable parameter shall be ≥ +13 dBm over the operating frequency range.

3.5.2.5.3 Minimum Level Output Range. The instrument shall have a minimum level output range of ≤ -15 dBm.

3.5.2.5.4 CW Mode Flatness and Accuracy. The instrument shall have the capability of meeting the following requirements of accuracy and flatness.

3.5.2.5.4.1 Flatness. The output shall be internally leveled for CW, step, or ramp modes, and shall be capable of the minimum acceptable parameter of ± 0.8 dB.

3.5.2.5.4.2 Accuracy. The output shall be internally leveled for CW, step, or ramp modes, and shall be capable of the minimum acceptable parameter of ±1.0 dB.

3.5.2.5.5 Analog Sweep Mode Flatness and Accuracy. The instrument shall be capable of an analog sweep mode flatness and accuracy over the frequency range of 0.01 to ≤0.05 GHz of ≥± 2.5 dB and for >0.05 to 20 GHz of ±4.0 dB.

3.5.3 Modulation.

3.5.3.1 Amplitude Modulation (AM). The instrument shall be capable of the following AM modulation specifications. The specifications shall be applied at ≥ 50% of modulation depth, at a rate of 1 kHz, with the RF set 6 dB below the maximum specified leveled output power.

3.5.3.1.1 AM Depth. The instrument shall be capable of an AM depth of modulation of 0 to 90%.

3.5.3.1.2 AM Bandwidth (3dB). The instrument shall be capable of an AM bandwidth of DC to 50 kHz.

3.5.3.1.3 Flatness (DC to 10 kHz rates). The instrument shall be capable of flatness of ≤ ±3 dB.

3.5.3.1.4 Accuracy. The instrument shall be capable of an accuracy of ≤ ±5%.

3.5.3.1.5 Distortion. The instrument shall be capable of a distortion of ≤ 5%.

3.5.3.1.6 External AM Input. The instrument shall have an external AM input capable of a log AM or linear AM input, with either a 50 Ω or 600 Ω input impedance.

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3.5.3.1.7 Sensitivity. The instrument shall have a sensitivity for Log AM that is continuously variable from 0 dB/volt to 25 dB/volt, and a Linear AM that is continuously variable from 0%/volt to 100 %/volt.

3.5.3.1.8 Maximum Input. The instrument shall be capable of a maximum input of ±1 volt.

3.5.3.1.9 Internal AM Generator.

3.5.3.1.9.1 Waveforms. The internal AM generator shall be capable of producing the waveforms of sinusoid, squarewave, triangle, positive ramp, negative ramp, Gaussian noise, and uniform noise.

3.5.3.1.9.2 Rate. The internal AM generator shall be capable of the rate of 0.1 Hz to 1 MHz for sinusoid and 0.1 Hz to 100 kHz for square wave, triangle, and ramps.

3.5.3.1.9.3 Resolution. The internal AM generator shall be capable of a resolution of 0.1 Hz.

3.5.3.1.9.4 Accuracy. The accuracy shall be the same as the time base.

3.5.4 Frequency Modulation (FM)

3.5.4.1 Maximum FM Deviation. The maximum FM deviation for the locked mode (1kHz to 8 MHz rates) shall be ± 10 MHz or less and for the locked low noise mode (50 kHz to 8 MHz rates) it shall be ± 10 MHz or less. The instrument shall have an unlocked wide mode at the (DC to 100 Hz rate) ±100 MHz.

3.5.4.2 FM Bandwidth (3dB). The FM 3dB bandwidth shall meet the following characteristics for Locked Mode of 1 kHz to 500 kHz, Locked Low Noise Mode of 30 kHz to 10 MHz, Unlocked Narrow Mode of DC to 500 kHz, and Unlocked Wide Mode of DC to 100 Hz.

3.5.4.3 Flatness locked mode (10 kHz to 500 kHz rates): ≤±1 dB.

3.5.4.3.1 Accuracy (100 kHz rate, ±1 V input). ≤10%.

3.5.4.3.2 External FM Input**.** The instrument shall have a 50 Ω or 600 Ω input impedance, which shall be selectable options selectable from modulation menu.

3.5.4.3.3 FM Sensitivity: The instrument shall have an FM sensitivity that is continuously variable from ±10 kHz per volt to ±20 MHz per volt (Locked and Unlocked Narrow FM modes), and ±100 kHz per volt to ±100 MHz per volt (Unlocked Wide FM mode), selectable from modulation menu.

3.5.4.3.4 Maximum Input. The instrument shall have a maximum input of ≥ ±1V.

3.5.5 Pulse Modulation (PM). The instrument shall provide PM functionality meeting the following specifications.

3.5.5.1 On/Off Ratio. The instrument shall be capable of an On/Off ration of ≥ 80 dB.

3.5.5.2 Rise/Fall Time. The instrument shall be capable of rise/fall times over the frequency range of 10 MHz to 1 GHz of ≤ 15 ns and over the frequency range of 1 GHz to 20 GHz of ≤ 10 ns.

3.5.5.3 Minimum Leveled Pulse Width. The instrument shall be capable of minimum level pulse widths of ≤ 100 ns for frequencies ≥ 2 GHz and ≤ 1 µs for frequencies < 2 GHz.

3.5.5.4 Minimum Unleveled Pulse Width. The instrument shall be capable of a minimum unleveled pulse width of ≤ 10 ns.

3.5.5.5 Pulse Overshoot. The instrument shall be capable of a pulse overshoot of ≤ 10 ns.

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3.5.5.6 Level Accuracy Relative to CW (100 Hz to 1 MHz PRF). The instrument shall be capable of level accuracies relative to CW of ±0.5 dB, for a pulse width ≥ 1 µs and ± 1.0 dB for pulse widths of < 1 µs.

3.5.5.7 PRF Range. The instrument shall be capable of the PRF range for the unleveled of DC to 10 MHz and 100 Hz to 5 MHz for the leveled.

3.5.5.8 External Input. The external input shall have a TTL compatible input and accept input logic of positive-true or negative-true.

3.5.5.9 Internal Pulse Generator. The internal pulse generator shall have the capabilities to provide the required pulse parameters stated herein.

3.5.5.10 Modes. The instrument shall be capable of the following modes: Free-run, triggered, gated, delayed, singlet, doublet, triplet, and quadruplet.

3.5.5.11 Accuracy. The instrument shall have a minimum accuracy of less than or equal to 10 ns.

3.6 Remote Operation. All instrument functions shall be accessible through an IEEE-488 interface bus (except for power on/standby).

3.7 Physical Parameters. The instrument shall have the following maximum dimensions and weight.

3.7.1 Dimensions. The width and depth of the equipment shall be compatible with mounting in an ANSI/EIA-310-D standard equipment rack. The height shall be compatible with four standard height units, 4U. One unit or "U" is defined by ANSI/EIA-310-D to be 1.75 inches or 44.45 mm.

3.7.2 Weight. The total weight of the equipment, excluding accessories and manuals, shall not exceed a two-person lift of 74 pounds (or 33.6 kilograms). If the depth of the equipment is greater than 24 inches the maximum allowable weight shall be reduced by 33 percent. If the depth of the equipment is greater than 36 inches the maximum allowable weight shall be reduced by 50 percent. If the depth of the equipment is greater than 48 inches the maximum allowable weight shall be reduced by 66%.

3.8 Manuals. The equipment shall be delivered with operator, maintenance, calibration, and illustrated parts manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, shall be as specified in the contract or order.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the products acceptable.

5.3 The contracting officer has the option of accepting or rejecting the product.

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6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Sources of Documents.

7.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

7.1.2 The Code of Federal Regulations, (CFR), may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

7.1.3 ANSI/EIA standards can be obtained from the Electronics Industry Association, Engineering Department, 2001 Pennsylvania Ave., N.W., Washington, D.C., 20006.
Phone: 800-854-7179 (USA and Canada).

MILITARY INTEREST

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